ALLIANCE UNIVERSITY | NAAC GRADE A+ ACCREDITED UNIVERSITY

Alliance School of Advanced Computing

**Mini Project Title: Mobile Penetration Test Report Generator with CVSS 3.1 Calculator**

**Course:** Mobile Penetration Testing & Ethical Hacking

**Course code:** 7CS1292


**Team Members:**

- Bachireddy Alekhya — 2023BCSE07AED083
- Tejashwini R — 2023BCSE07AED208
- Srinivas R — 2023LCSE07AED659

**Faculty:** Dr. Durgadevi




# 02- Problem Statement & Objectives

**Problem Statement**

Penetration-testing reports require vulnerability details, CVSS scores, evidence and remediation information. Managing these manually can be time-consuming and inconsistent.

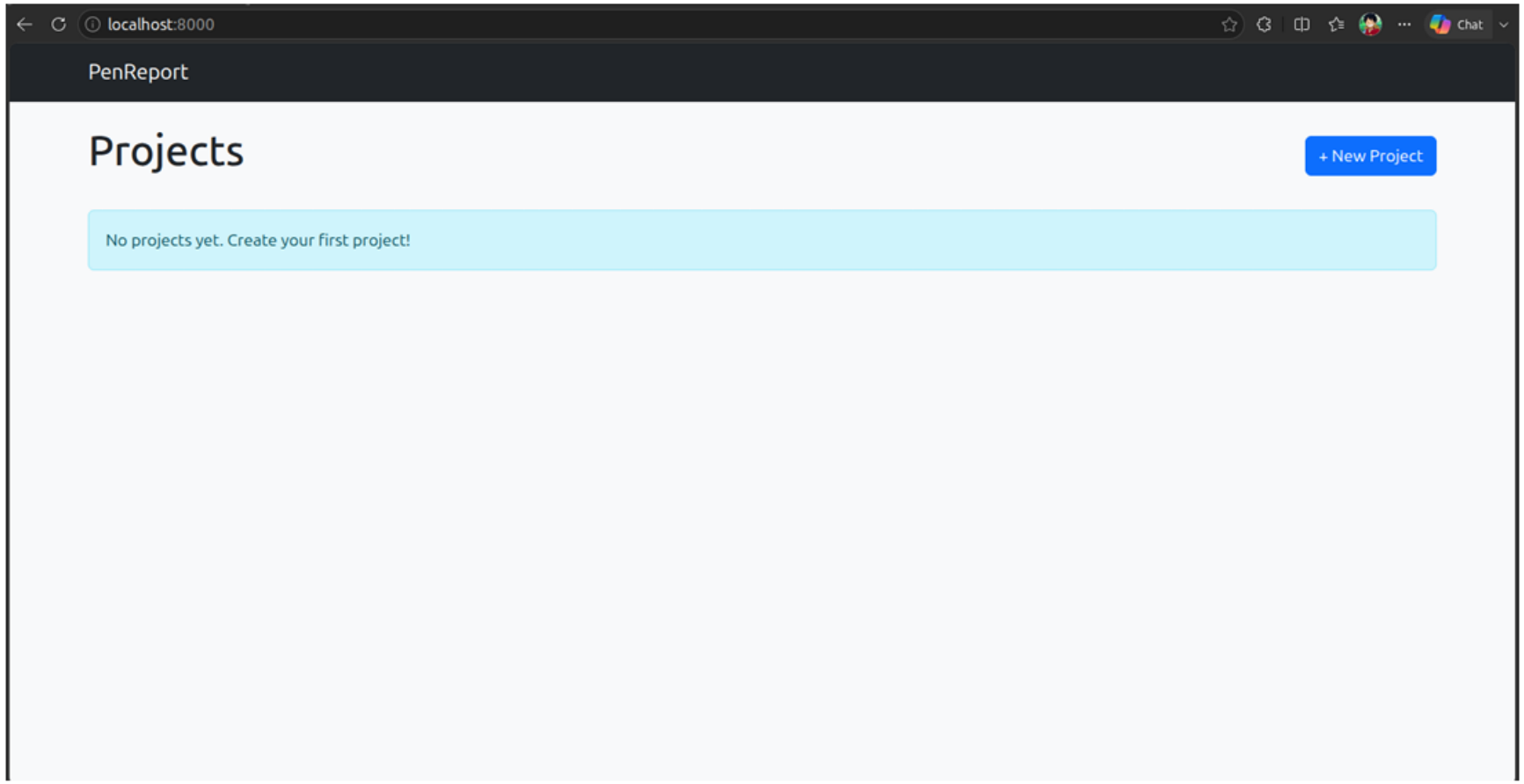
**Why is it important?**

A single platform can reduce repetitive documentation work and maintain consistent vulnerability scoring and report formatting.

**Objectives**

- Develop a simple web-based reporting platform.
- Integrate CVSS 3.1 scoring with severity feedback.
- Support evidence attachment for findings.
- Generate structured PDF security reports.

# Dashboard


localhost:8000
Chat
PenReport
Projects
+ New Project
No projects yet. Create your first project!

# 03- REQUIREMENTS

**Functional Requirements**

- The system allows the tester to create projects, manage findings, calculate CVSS scores, upload evidence and generate reports.
- Project & finding management
- CVSS 3.1 calculation
- Evidence upload
- PDF report generation

**Non-Functional Requirements**

- The application should be simple, consistent and suitable for local use.
- User-friendly interface
- Consistent scoring
- Local operation
- Readable report output

**Hardware:** Dual-core CPU, 2 GB+ RAM, ~100 MB storage
**Software:** Python 3.10+, FastAPI, SQLite, Chrome/Firefox/Edge

# 04- PROPOSED SOLUTION

**PenReport**

PenReport is a web-based penetration-testing report management system that combines vulnerability documentation, CVSS scoring, evidence management and automated PDF generation in one platform.

**Key Features**

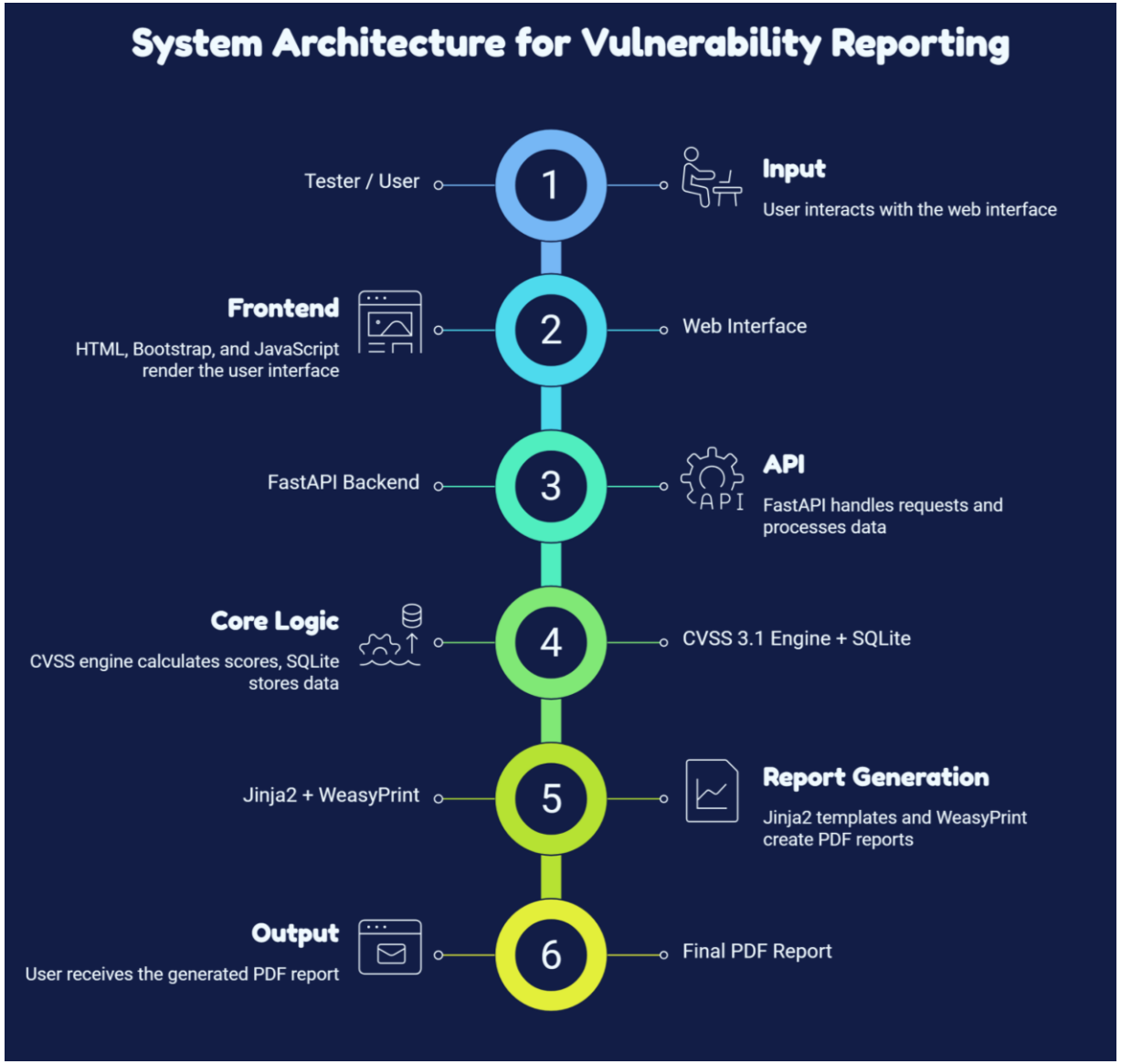
Project Management → Create and manage testing projects
Finding Management → Record vulnerability and remediation details
CVSS Calculator → Calculate score and severity
Evidence Management → Attach supporting screenshots
PDF Generation → Create structured security reports

**Advantage**

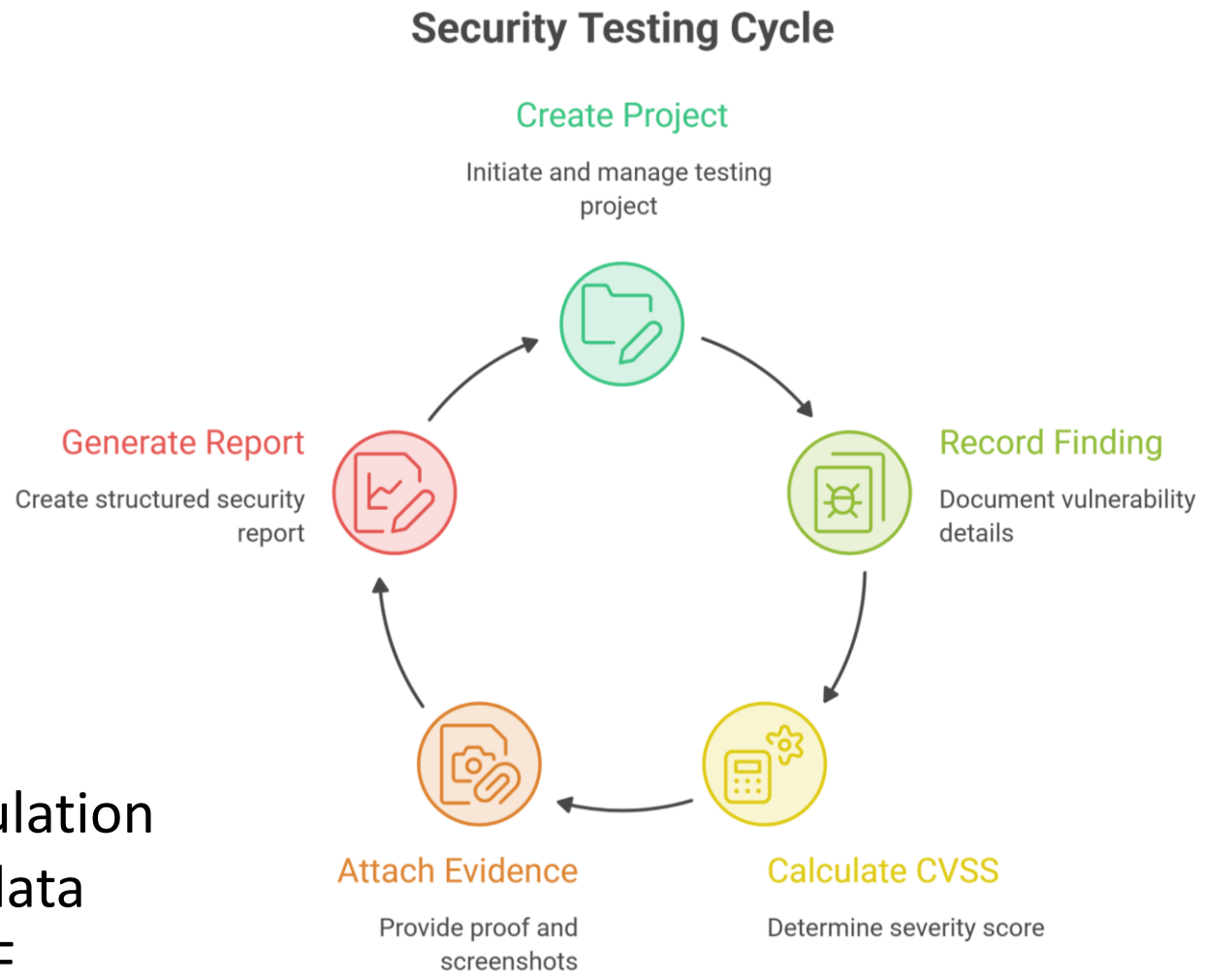
**Instead of managing findings, scoring, evidence and reporting separately, PenReport provides them through one integrated workflow.**

**Project → Finding → CVSS → Evidence → PDF**

# 05- ARCHITECTURE / WORKFLOW



# 05- ARCHITECTURE / WORKFLOW




## Major Components

- **Frontend:** User interaction
- **FastAPI:** Application logic
- **CVSS Engine:** Score & severity calculation
- **SQLite:** Stores project and finding data
- **WeasyPrint:** Converts report to PDF

# 06- TECHNOLOGIES & IMPLEMENTATION

| Category | Technologies |
|---|---|
| Language | Python |
| Backend | FastAPI, Uvicorn |
| Database | SQLite, SQLAlchemy |
| Frontend | HTML, CSS, Bootstrap 5, JavaScript |
| Reporting | Jinja2, WeasyPrint |
| Security | CVSS 3.1 |

**Implementation**

The system follows a simple pipeline:

Project Creation → Finding Management → CVSS Calculation → Database Storage → Evidence → PDF Generation → Testing

# 07- RESULTS & APPLICATIONS

**Results**

The prototype was tested for its major functions and the recorded tests passed successfully.
Project Creation | Finding Management | CVSS Scoring | Evidence Upload | PDF Generation

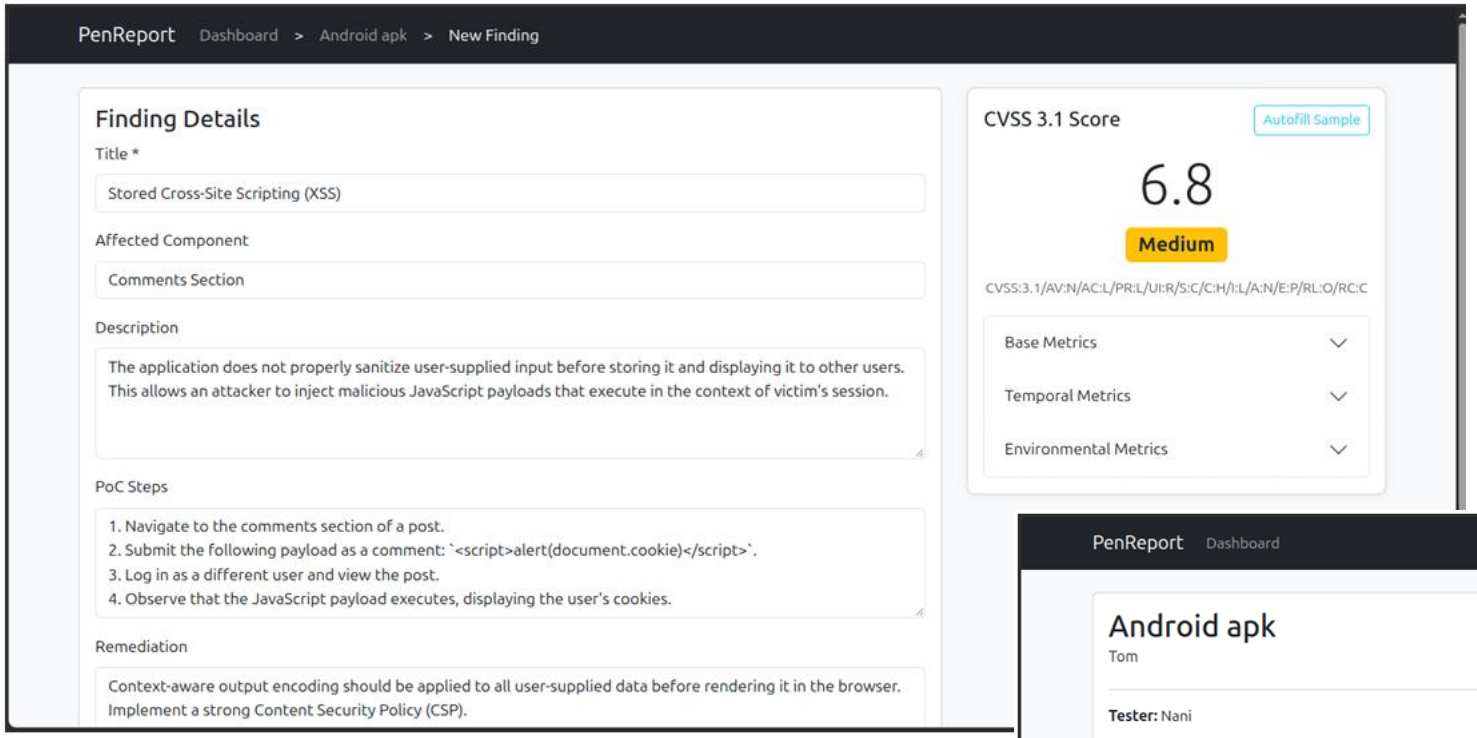
**Applications**

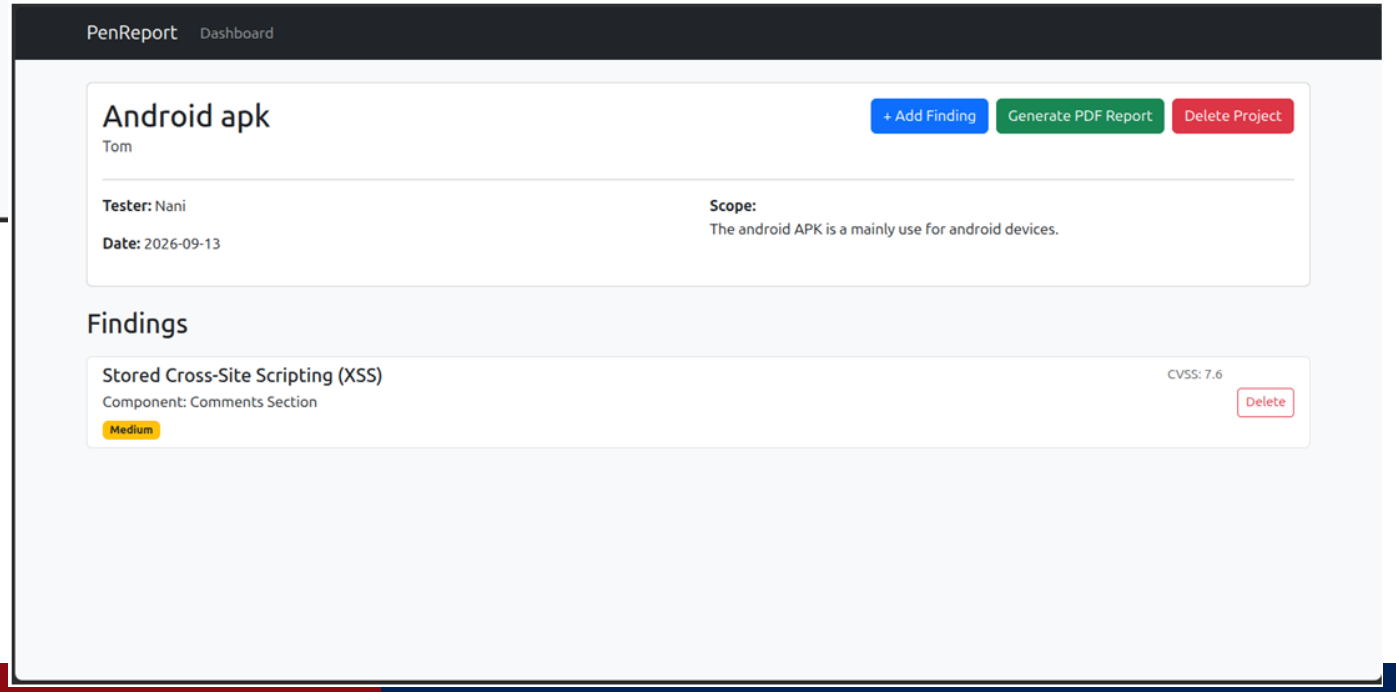
- Penetration-testing reporting
- Mobile application security testing
- Vulnerability assessment
- Security audit documentation

**Challenges & Solutions**

| Challenge | Solution |
|---|---|
| CVSS calculation | Integrated CVSS 3.1 calculator |
| Evidence management | Finding-based screenshot upload |
| Report formatting | Automated PDF generation |

# 07- RESULTS & APPLICATIONS



# 08- CONCLUSION & FUTURE SCOPE

**Conclusion**

PenReport provides a simple and organized way to document vulnerabilities, calculate CVSS 3.1 scores, manage evidence and generate structured PDF reports.

**Key Outcomes**

Reduced manual reporting effort
Consistent vulnerability scoring
Organized evidence and findings
Automated report generation

**Future Scope**

User authentication
JSON/CSV report formats
Vulnerability dashboards
CVSS 4.0 support
Cloud & multi-user deployment
Docker deployment

# 09- References


**1. Penetration Frameworks and Development Issues in Secure Mobile Application Development: A Systematic Literature Rev**
**IEEE Access, 2021**
**Note:** Explains common frameworks and techniques used for mobile application penetration testing.
**Link:** IEEE / DOI
**2. Dynamic Security Analysis on Android: A Systematic Literature Review**
**IEEE Access, 2024**
**Note:** Reviews techniques used to identify security vulnerabilities in Android applications. Orbilu
**Link:** IEEE / DOI
**3. Static Analysis of Android Apps: A Systematic Literature Review**
**Information and Software Technology, Elsevier, 2017**
**Note:** Explains how static analysis can be used to detect security and privacy issues in Android apps. ScienceDirect
**Link:** ScienceDirect / DOI
**4. Common Vulnerability Scoring System**
**IEEE Security & Privacy, 2006**
**Note:** Introduces CVSS as a standard method for measuring vulnerability severity. NIST
**Link:** NIST / Paper
**5. Can the Common Vulnerability Scoring System be Trusted? A Bayesian Analysis**
**IEEE Transactions on Dependable and Secure Computing, 2018**
**Note:** Studies the reliability and consistency of CVSS vulnerability scores. IEEE Xplore
**Link:** IEEE Xplore / Paper

ALLIANCE® UNIVERSITY | NAAC GRADE A+ ACCREDITED UNIVERSITY

Department of Computer Science and Engineering

Alliance School of Advanced Computing

# THANK YOU

©ASAC – Alliance University, 2026. All rights reserved. Unauthorized reproduction or distribution is prohibited.